Socratic Skill Learning: Interleaving Teaching and Practice for Transferable Agent Skills

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HarnessRSI 69 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Educator
stronger model, READ-ONLY on recent work and the live package: explain, challenge, diagnose, or stay silent

Student
the solver's own model: sole environment operator and sole Skill author; may question, test, adapt, reject advice

When does the Educator review?

Fixed-Window
after every g tool calls (g = 10 main; 5, 20 tested), plus a final review each round

Issue-Guided
after a first review, only when a learning issue opens, changes or reopens

Acquisition: 3 rounds on ONE anchor task + public spec

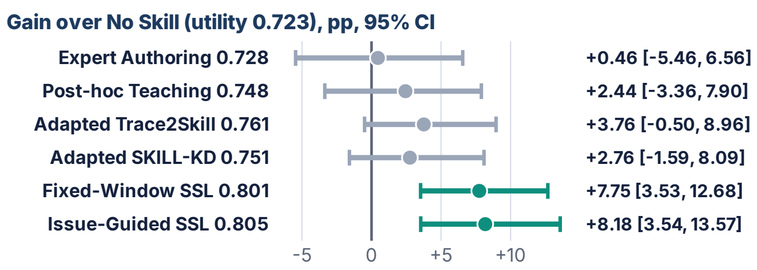
1. **Practice + review** — Student explores the anchor, edits SKILL.md / scripts
2. **Fresh solver tests the package** — on the reset anchor; verifier score returned
3. **Feedback to next round** — trajectory + score; contradiction can reopen an issue

×3

Final package frozen → fresh solver per held-out case of the same family (private tests)
weights never change; no best-round selection described

Changed artifact: an Agent Skill package (SKILL.md + scripts), not weights. Benchmark SkillTransferBench: 35 families, 408 cases; main cohort 22 families, 155 held-out cases.

2 Headline: 0.801 / 0.805 vs 0.723 no Skill



Qwen3.7 Plus student, Max Educator; mean of 3 repeats. **Interval crossing 0 = not separated from No Skill.** Fixed leads Trace2Skill by 3.99 [1.45, 6.49].

4 Where the gain is absent or within noise

- No baseline CI clears zero; Expert vs Post-hoc order flips between cohorts, within noise
- **Issue vs Fixed: 0.805 vs 0.801**; no main-cohort interval, repeat SD 0.030 / 0.036. DeepSeek 0.779 vs 0.784, Kimi 0.753 vs 0.762: similar utility
- Fixed improves 16 of 22 families, **4 fall** (Bank filter 0.678 → 0.593, SPARQL 0.706 → 0.680, Fin. docs, XLSX). Four families give ≈ 5.0‡ of the 7.75 pp, not SSL-specific: DB WAL 0.473 → 0.884, Trace2Skill 0.905
- Case-study family (HTML finance): Expert 1.000 vs Fixed 0.917

‡ computed by us from Table 9.

3 Controls that decide if the gain is real

Same solver, cases, verifier for all methods

Expert / Post-hoc / SSL: 3 rounds + anchor feedback; under Expert the Max model writes the package. Trace2Skill / SKILL-KD: author-adapted, no such loop, cost not on the same scope.

Review schedule (Table 17)

Qwen main · 3 repeats	utility	tokens/run	reviews
Fixed g = 5	0.776	9.56 M	1126
Fixed g = 10	**0.801**	6.89 M	630
Fixed g = 20	0.776	6.61 M	375
Issue-Guided	**0.805**	5.72 M	329

Why g = 10 is main is not stated; utility non-monotonic in g. Issue-Guided: 60.6 M vs 158.2 M Educator tokens (61.7% fewer), with trigger, context and follow-up changed together.

Three model pairs, 10 families / 77 cases (Table 2a)

Gain, pp (lite)	Fixed	Issue	Expert	Post-hoc
Qwen (3 rep.)	+4.1	+4.5	+1.1	**−0.7**
DeepSeek (1)	+3.1	+2.5	+1.2	+0.5
Kimi (1)	+8.2	+7.4	+5.8	+4.1

Fixed minus No Skill: DeepSeek [−4.63, +13.03] includes 0. Fixed minus Expert and Issue minus Fixed: all three CIs include 0.

Scope

Observational lesson → edit link (112 of 193 windows). Cohort picked by No-Skill screening; overlap with reported No-Skill runs not stated.

Takeaway

Reviewing while the model practises, with the model as sole Skill author, scores highest on the main cohort, but only the two SSL schedules have gains whose CI clears zero. Whether issue-guided review beats fixed windows on utility is not shown; it spends fewer Educator tokens at similar utility.

Our read · HarnessRSI

Score 69 · ★★ · evidence level: retained

Reports: held-out cases, repeats + CIs, 3 model pairs, costs.
Does not report: matched-token runs, contamination check, cross-family transfer, baseline tuning budgets.